

NEET CHEMISTRY

Preparation Quick Notes & Revision Guide

Physical + Inorganic + Organic Chemistry

This original revision guide is designed for NEET preparation and follows the broad Chemistry areas in the official NEET (UG)-2026 syllabus published by NTA/NMC.

Section	Focus
Physical Chemistry	Formulae, concepts and numericals
Inorganic Chemistry	Periodic trends, blocks and coordination
Organic Chemistry	GOC, hydrocarbons and functional groups
Practice	MCQs and a 7-day revision plan

Official reference: NTA's NEET (UG)-2026 syllabus notice.

1. Physical Chemistry - High-Yield Revision

Some Basic Concepts of Chemistry

Use $n = \text{mass} / \text{molar mass}$. Number of particles = $n \times N(A)$. Keep units consistent for molarity, molality and mole fraction.

Atomic Structure

Revise quantum numbers, electronic configuration, Aufbau principle, Pauli exclusion principle and Hund's rule. Remember $c = \nu \times \lambda$ and $E = h \times \nu$.

Chemical Bonding

Focus on Lewis structures, formal charge, VSEPR, hybridisation and basic molecular-orbital ideas. sp is linear, sp^2 is trigonal planar and ideal sp^3 is tetrahedral.

Thermodynamics

Revise system and surroundings, enthalpy, Hess's law, entropy and Gibbs energy. $\Delta G = \Delta H - T \Delta S$.

Equilibrium

Know K_c , K_p , Le Chatelier's principle, ionic equilibrium, pH, buffers and solubility product.

Electrochemistry

Revise galvanic/electrolytic cells, electrode potential, Nernst equation, conductance and Faraday's laws. $E_{\text{cell}} = E_{\text{cathode}} - E_{\text{anode}}$.

Chemical Kinetics

Focus on rate law, order, integrated equations, half-life and Arrhenius equation. For first order: $t(1/2) = 0.693/k$.

Solutions

Revise concentration terms, Raoult's law and colligative properties including osmotic pressure.

2. Inorganic Chemistry - High-Yield Revision

Periodicity

Across a period, atomic radius generally decreases while ionisation enthalpy and electronegativity generally increase. Down a group, atomic size generally increases.

p-Block Elements

Revise Groups 13-18, electronic configuration, oxidation states, trends, anomalous behaviour and important compounds.

d- and f-Block Elements

Focus on transition-element configurations, oxidation states, colour, magnetic behaviour, catalysis, complexes and alloys. Know important chemistry of KMnO_4 and $\text{K}_2\text{Cr}_2\text{O}_7$.

Coordination Compounds

Know ligand, coordination number, denticity, chelation, oxidation state, nomenclature and isomerism. Revise basic crystal-field ideas, colour and magnetism.

Study Method

Use NCERT as the primary factual revision source. Make one-page tables for trends, oxidation states, colours, exceptions and important compounds.

Fast Recall Checklist

Topic	Check
Periodicity	Can I explain major trends and their reasons?
p-Block	Can I recall major trends and important compounds?
Transition elements	Can I explain colour, magnetism and variable oxidation states?
Coordination	Can I identify ligands and coordination number and name simple complexes?

3. Organic Chemistry - High-Yield Revision

General Organic Chemistry

Master tetravalency, hybridisation, functional groups, IUPAC nomenclature, structural/stereo-isomerism and common reaction types. Understand electrophiles, nucleophiles, carbocations, carbanions and free radicals.

Electronic Effects

Revise inductive effect, electromeric effect, resonance and hyperconjugation. Use them to understand stability, acidity/basicity and reaction behaviour.

Hydrocarbons

Revise preparation, properties and important reactions of alkanes, alkenes, alkynes and aromatic hydrocarbons.

Haloalkanes and Haloarenes

Focus on C-X bond polarity, substitution, elimination and reaction conditions.

Alcohols, Phenols and Ethers

Revise preparation, acidity of phenols, dehydration of alcohols and important reactions.

Aldehydes, Ketones and Carboxylic Acids

Focus on carbonyl polarity, nucleophilic addition, oxidation/reduction, acidity and important reactions/tests.

Amines and Biomolecules

Revise classification and basicity of amines, important reactions, and key facts about carbohydrates, proteins, enzymes, vitamins and nucleic acids.

Organic Strategy

Learn reactions as substrate → reagent/condition → product → reaction type. Keep a separate reaction sheet and revise it regularly.

4. Quick NEET Chemistry Practice

Q1. Under standard conditions, E-cell is calculated as:

- A. E-anode - E-cathode B. E-cathode - E-anode C. E-anode + E-cathode D. Always zero

Q2. The half-life of a first-order reaction is:

- A. $1/k$ B. $0.693/k$ C. $k/0.693$ D. $2k$

Q3. Ideal geometry for sp^3 hybridisation is:

- A. Linear B. Trigonal planar C. Tetrahedral D. Octahedral

Q4. Which concept helps explain the stability of many organic intermediates?

- A. Resonance B. Only density C. Only boiling point D. Only molecular mass

Q5. Which pair belongs to coordination chemistry?

- A. Ligand and coordination number B. Osmosis and molality C. Half-life and rate constant D. Hess's law and entropy

Answers: 1-B, 2-B, 3-C, 4-A, 5-A

5. 7-Day Chemistry Revision Plan

Day	Focus
1	Mole concept + atomic structure + chemical bonding
2	Thermodynamics + equilibrium
3	Solutions + electrochemistry + kinetics
4	Periodicity + p-block
5	d/f-block + coordination compounds
6	GOC + hydrocarbons + functional groups
7	Full revision + mixed MCQs + error notebook

Revision rule: After every chapter, solve MCQs, record mistakes in an error notebook and revisit weak concepts after 24-48 hours.

The official NEET (UG)-2026 Chemistry syllabus includes Physical, Inorganic and Organic Chemistry units; this guide is a concise original revision aid, not a replacement for the official syllabus or NCERT.